Executive Summary

Toronto Island Ferry ticket sales and redemptions, May 2015 to December 2025

Prepared 18 October 2026 for Toronto stakeholders. Every figure is calculated from the supplied dataset (261,538 fifteen-minute records).

Business problem

Ferry service to Toronto Island Park needs to match staffing and sailings to a highly variable demand. This analysis uses 15-minute ticket sales and redemption counts to show when demand peaks, how much it varies by season and weekday, and how far the data can be trusted.

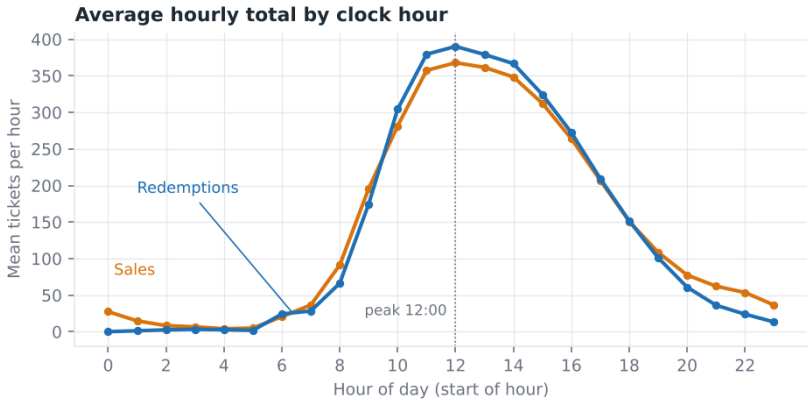
At a glance	
Period covered	2015-05-01 to 2025-12-21 (3,882 days with data)
Tickets redeemed / sold	12,785,293 / 12,972,051
Busiest hour	12:00-13:00 (390 redemptions per hour on average)
Busiest period	11:00-14:00 carries 35% of each day's redemptions
Busiest season / month	Summer; August (9,908 redemptions per day)
Quietest season / month	Winter; January (492 per day)
Weekend vs weekday	1.88x overall; 1.91x in summer; 0.99x in winter
Data completeness	80% of 15-minute slots recorded between each day's first and last record; the rest appear to be intervals with no activity

Dataset coverage and quality

The file is clean: no missing values, negative counts or duplicate records, and all timestamps sit on 15-minute boundaries. Six calendar days have no records, and the file omits intervals with no activity, so 63,951 slots between each day's first and last record are absent. 472 unusually high intervals were flagged and kept. This is **historical data** ending 2025-12-21; it is not a live feed.

Most important verified findings

- **Demand is concentrated in summer.** June to August holds 63% of all redemptions; December to February holds 4%. A winter day averages 7% of a summer day.
- **Demand is concentrated in the middle of the day.** Hours 10:00 to 15:59 account for 65% of redemptions. Saturday at 13:00 is the single busiest weekday-hour.
- **Weekends matter only in the warm season.** Summer weekends average 1.91 times summer weekdays; in winter weekends and weekdays are the same.
- **Two years were exceptional.** Redemptions in May to July 2017 were 93% below the normal level for those months, and April to September 2020 were 78% below. The data do not say why.
- **2025 was the busiest year in the file** at 1,519,276 redemptions, even though the data end on 21 December.
- **Sales and redemptions track each other** (daily correlation 0.98; 1.5% more tickets sold than redeemed overall), but sales continue overnight when almost no redemptions occur, so they must not be read as passenger movements.



Average tickets per clock hour. Redemptions peak at midday; sales stay higher late at night.

Peak and off-peak periods

	Peak	Off-peak
Hour	12:00-13:00 (390 per hour)	23:00-24:00 (13.5 per hour); overnight is near zero
Day	Saturday (5,036 per day)	Tuesday (2,290 per day)
Month	August (9,908 per day)	January (492 per day)
Season	Summer (index 240)	Winter (index 17)

Season index: 100 equals the average day of the whole period.

Operational implications

Demand is concentrated in summer, on weekends and around midday, while winter volumes are a small fraction of that. The heaviest sustained load found was about 3,132 redemptions per hour over four hours (July 2018). The data hold no vessel capacity or queue information, so whether this level strains operations is not known.

Recommended actions (to be validated by operations)

- Concentrate staffing, ticketing and queue management in the 11:00-14:00 window, with the heaviest cover on summer weekends.
- Plan service in three tiers (summer, spring/fall, winter) and assess whether a reduced winter pattern fits service obligations.
- Use 4-hour rolling redemption demand, not single 15-minute values, as the trigger for extra capacity.
- Confirm with operations what caused the 2017 and 2020 downturns and the extreme single-interval spikes before using those records for planning.

Limitations and next steps

- Counts are tickets, not passengers. The data have no route, capacity, queue, weather or event information.
- Findings are descriptive; recommendations are hypotheses, not tested conclusions.
- Next: add capacity, queue, route and weather data; define the two ticket measures formally; and connect a verified live source if real-time monitoring is required. An interactive dashboard (*streamlit run app.py*) lets staff explore every figure above.